SQL
S2

EDC- 2026
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SQL Class Rules

1. **It’s easier to learn it now and here!**
(Stay focused and ask questions.)

1. **AI is your assistant, not your brain!**
(Don’t let AI think instead of you.)

Course content

S1: Introduce the course, data concepts and the implementation of the software

S2: Create the database and its tables on the software

S3: Manipulate, retrieve and analyze data

S4: Establish the basic notions of using SQL

S5: Understand and practice jointures

S6: Settle the basics of SQL, Compare joins with subqueries

S7: Go further with the window functions

S8: Present the group work, Prepare for the exam

Session 2 :DDL & DML

- DDL & DML

- Import the 10 given datasets that will be used for the course

Attention

Your keyboard should be in English because:

- SQL uses English keywords and symbols (=, ;, ', etc.).
- Non-English layouts may cause syntax errors or misplaced characters.

Query Basics:

- A **query**: is a request for data from a database

- **Result set:** Results of our query.

- We **end** the query with a **semicolon** to indicate that the query is complete.

- **keywords are capitalized,** and **table and field names are all in lowercase.**

Overview of a Query:

```
CREATE TABLE table_name (
  id SERIAL          PRIMARY KEY,      -- Auto-incrementing ID
  column_one         TEXT,             -- A text column
  column_two         INTEGER           -- An integer column
);
```

Writing comment for a query: -- comment

Write a Query:

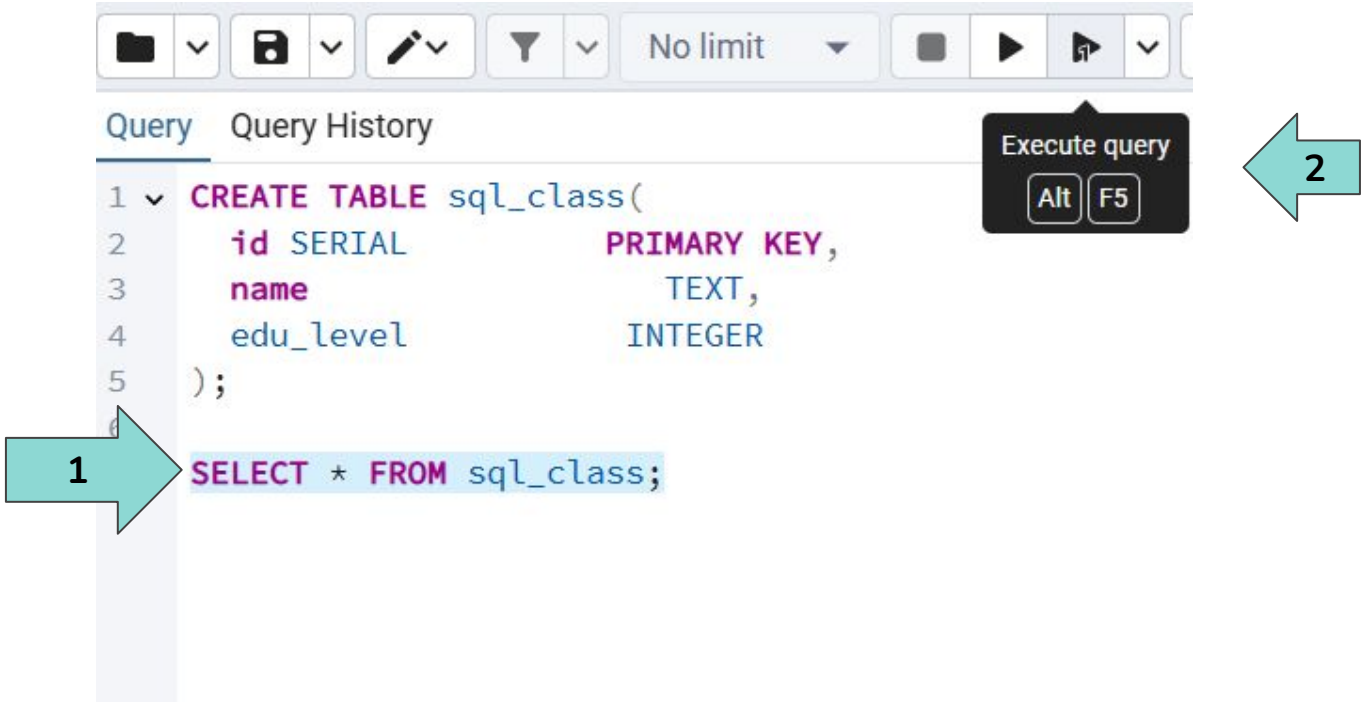
```
CREATE TABLE sql_class(
  id SERIAL          PRIMARY KEY,
  name               TEXT,
  edu_level          INTEGER
);
```

Execute the query:

1. Highlight the code you want to run.

2. Click the “Execute Query” button (or press Alt + F5).

Execute the query:



SQL is the language for interacting with **databases**.
It has several subsets:

1. DDL (Data Definition Language) defines or modifies the database **structure** (e.g., CREATE, ALTER, DROP).

2. DML (Data Manipulation Language) is used for managing **data** within the database (e.g., SELECT, INSERT, UPDATE, DELETE).

DDL handles structure, and DML handles data, both using SQL.

DDL (Data Definition Language):

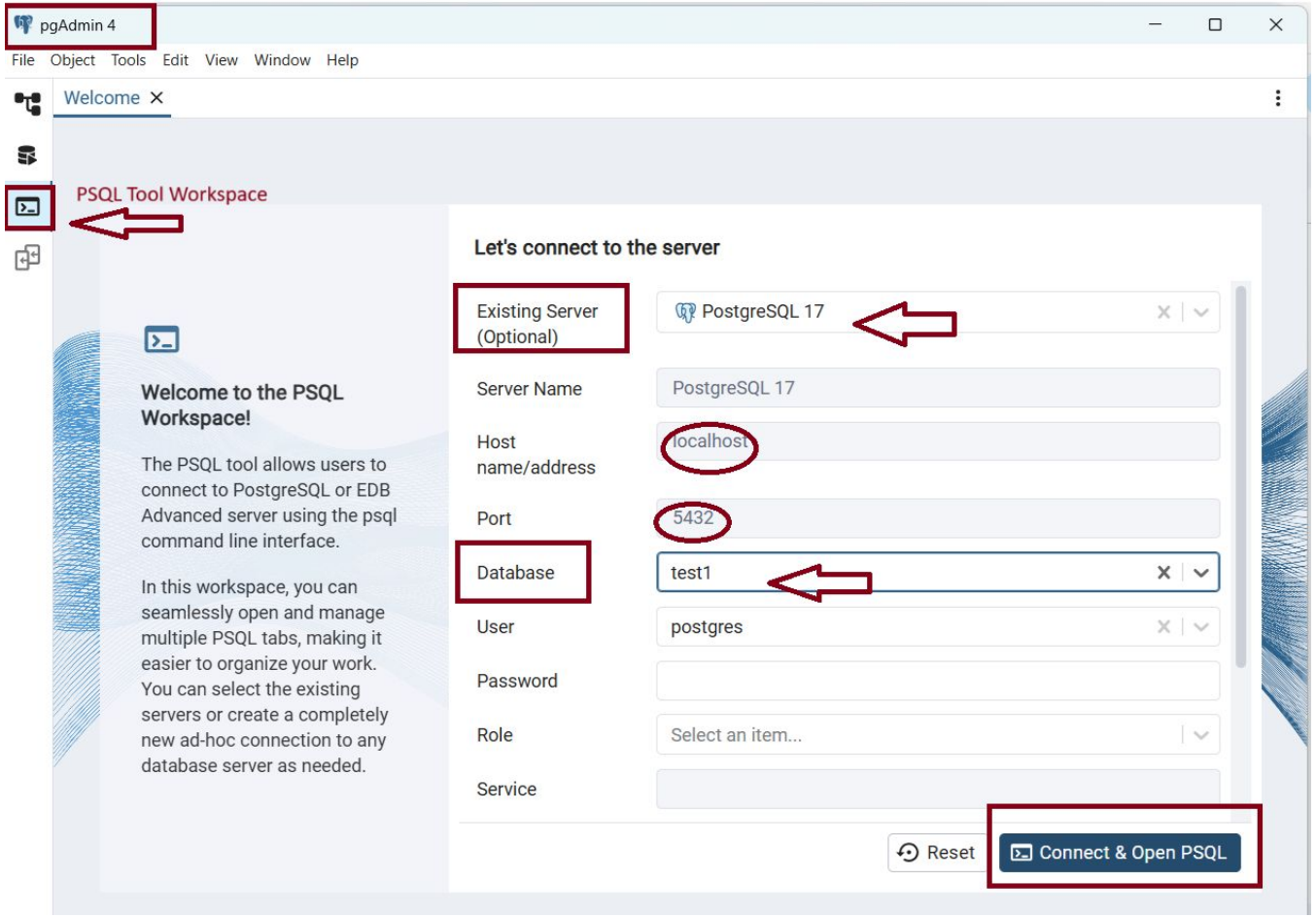
CREATE, DROP, ALTER, TRUNCATE, COMMENT, RENAME

Create a Database:

CREATE DATABASE test1;

test1=> Always use lowercase names and avoid spaces or special characters.

Refresh=> connect to db


pgAdmin 4
File Object Tools Edit View Window Help
Welcome
PSQL Tool Workspace
Welcome to the PSQL Workspace!
The PSQL tool allows users to connect to PostgreSQL or EDB Advanced server using the psql command line interface.
In this workspace, you can seamlessly open and manage multiple PSQL tabs, making it easier to organize your work. You can select the existing servers or create a completely new ad-hoc connection to any database server as needed.
Let's connect to the server
Existing Server (Optional)
PostgreSQL 17
Server Name
PostgreSQL 17
Host name/address
localhost
Port
5432
Database
test1
User
postgres
Password
Role
Select an item...
Service
Reset
Connect & Open PSQL

Example- CREATE

Create a table that shows the SQL class students’ ID, Name, and Grade.

CREATE

```
CREATE TABLE sql_class (
  id            INTEGER,
  name          TEXT,
  grade         INTEGER
);
```

To see the Result Set: ***SELECT * FROM sql_class;***

Example- Add Column=> ALTER

Add a “Participation” Column to the SQL Class Table.

Add Column=> ALTER

```
ALTER TABLE sql_class
ADD COLUMN participation VARCHAR(1);
```

Example- Change data type=> ALTER

Change the “ID” Column’s Data Type from INT to VARCHAR(2) to Store 2 Characters.

Change Data Type =>ALTER

```
ALTER TABLE sql_class
ALTER COLUMN id TYPE VARCHAR(2);
```

This command modifies the data type of the id column in the sql_class table.

Example- Rename To=> ALTER

Rename:

- The Table name to sql_students
- The “id” Column name to “key”

RENAME TO => ALTER

- ALTER TABLE sql_class **RENAME TO** sql_students**;**
- **ALTER TABLE** sql_students **RENAME** COLUMN id **TO** key;

SELECT * FROM sql_students;

TRUNCATE : Delete all data from the table.

TRUNCATE TABLE table_name;

DROP: Delete the table

DROP TABLE table_name;

DML (Data Manipulation Language):

Manage data inside the tables.

Examples: SELECT, INSERT, UPDATE, DELETE

INSERT : Add a new record

```
INSERT INTO sql_students (key, name, grade, participation)
VALUES
  (1, 'Jane', 19 , 'A'),
  (2, 'John', 18 ,'B' ),
  (3, 'Jack', 17, 'A')
;
```

SELECT: Get data from the table.

SELECT * FROM table_name;

SELECT column_name FROM table_name;

UPDATE: Change data in the table.

Change Jack’s grade to 20:

```
UPDATE sql_students
SET grade = 20
WHERE name = 'Jack';
```

```
SELECT name, grade FROM sql_students;
```

DELETE: Remove a row

```
DELETE FROM sql_students
WHERE  participation = 'B';
```

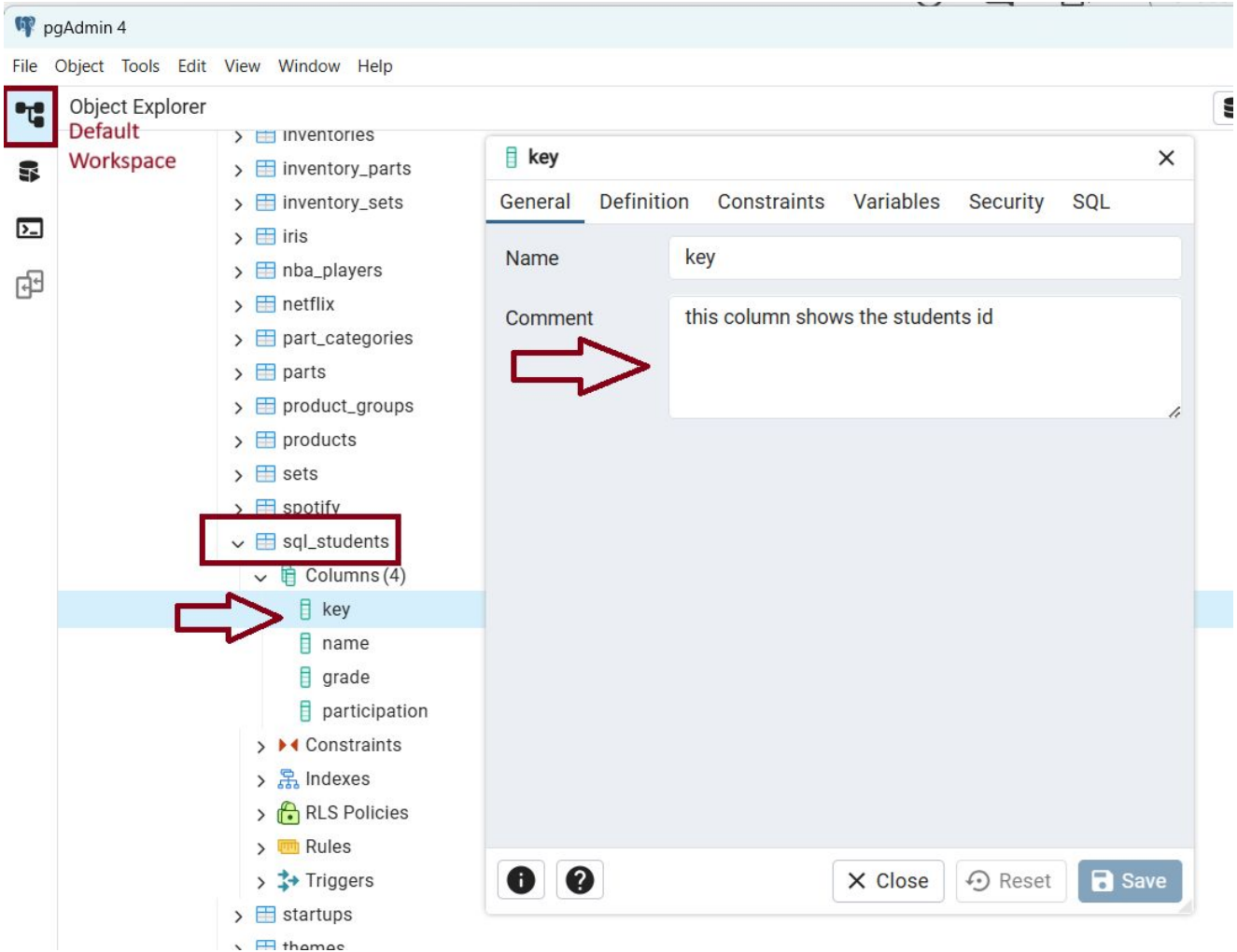
Use DELETE when you want to remove certain rows.

Use TRUNCATE to clear the whole table fast.

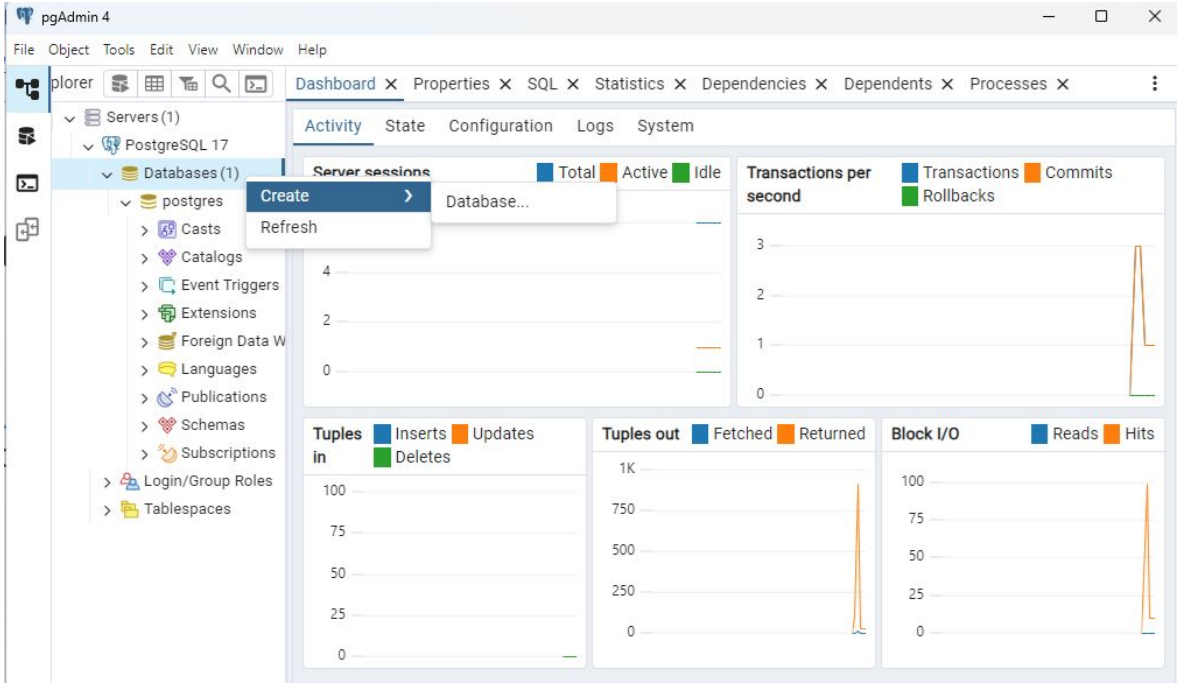
Use DROP when you no longer need the table at all.

COMMENT : Add a note about a column (DDL)

```
COMMENT ON COLUMN table_name.column_name IS
'This column is about...';
```


pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
Default Workspace
inventories
inventory_parts
inventory_sets
iris
nba_players
netflix
part_categories
parts
product_groups
products
sets
spotify
sql_students
Columns (4)
key
name
grade
participation
Constraints
Indexes
RLS Policies
Rules
Triggers
startups
key
General Definition Constraints Variables Security SQL
Name
key
Comment
this column shows the students id
Close
Reset
Save

Practice:Create a Database Using the UI

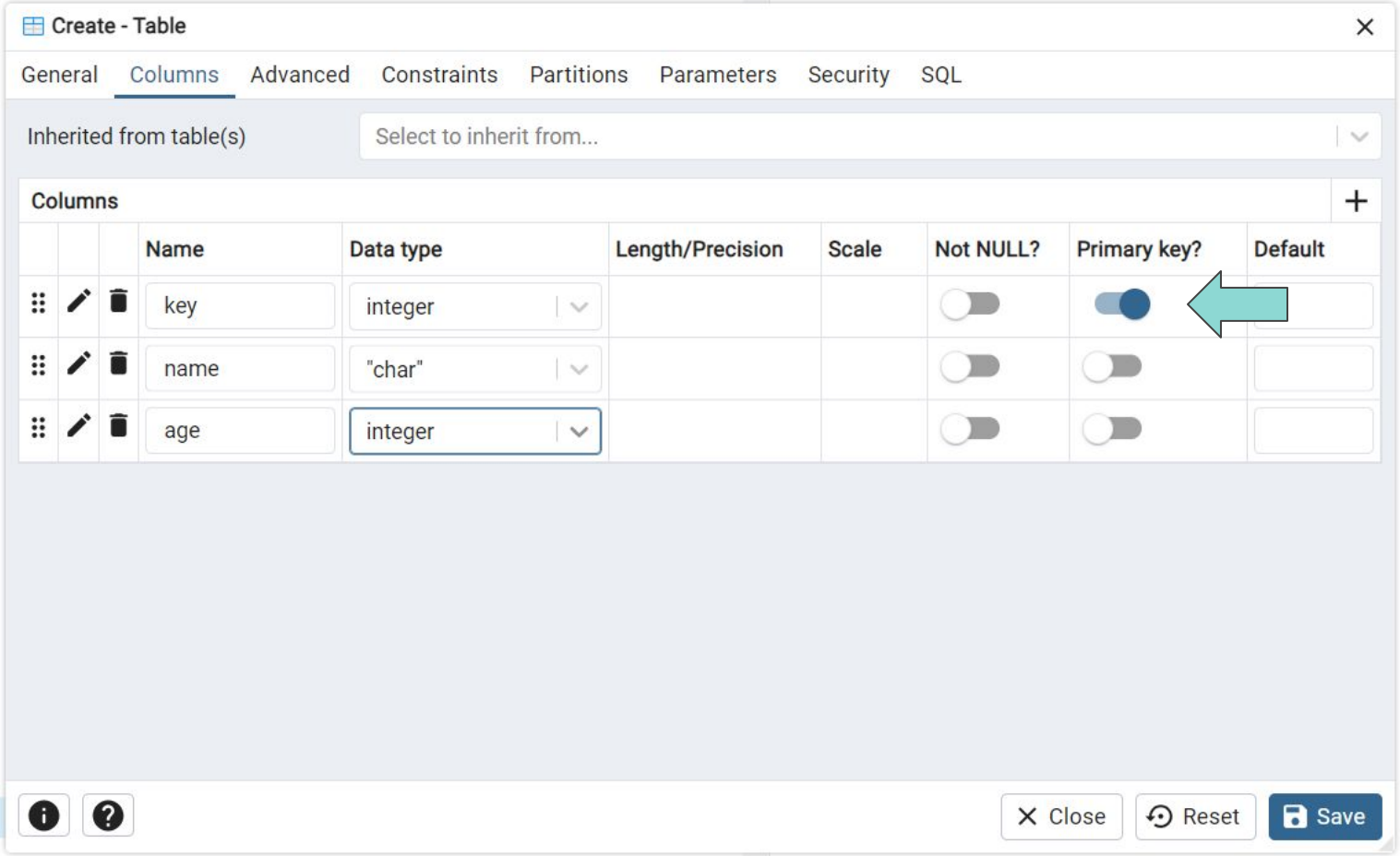


Practice: Create a Table Using the UI

Database: test1 => Tables=> Create Table=>

Name: users_ui

Columns: key, name, age


Create - Table
General
Columns
Advanced
Constraints
Partitions
Parameters
Security
SQL
Inherited from table(s)
Select to inherit from...
Columns
Name
Data type
Length/Precision
Scale
Not NULL?
Primary key?
Default
key
integer
name
"char"
age
integer
Close
Reset
Save

Column names must:

- Start with a letter (not a number).
- Contain only letters, numbers, and underscores.

Not use reserved keywords (like SELECT, FROM, etc.).

Avoid names like name!, 123score, select.

Data Types Used:

- VARCHAR(n) → Variable-length string, max n characters.
  Example: VARCHAR(50) stores up to 50 characters.

- INTEGER → Stores whole numbers.
  Example: confidence INTEGER can store numbers like 5, 10, etc.

- TEXT → Unlimited-length string. Use it when you don't want to restrict size (good for long answers like strengths).

Prepare for Dataset Import

Before importing:

- Open the CSV file
- Understand its schema (column names and types)
- Check for missing values or strange characters
- Match CSV columns to your table definitions

Import .CSV rules:

- The **number of columns** in the CSV file must match the table definition.
- The **order of columns** in the CSV must be the same as in the table.
- The **data types** in the CSV should match the data types defined in the PostgreSQL table.

Practice: Import .CSV

```
CREATE TABLE student_info (

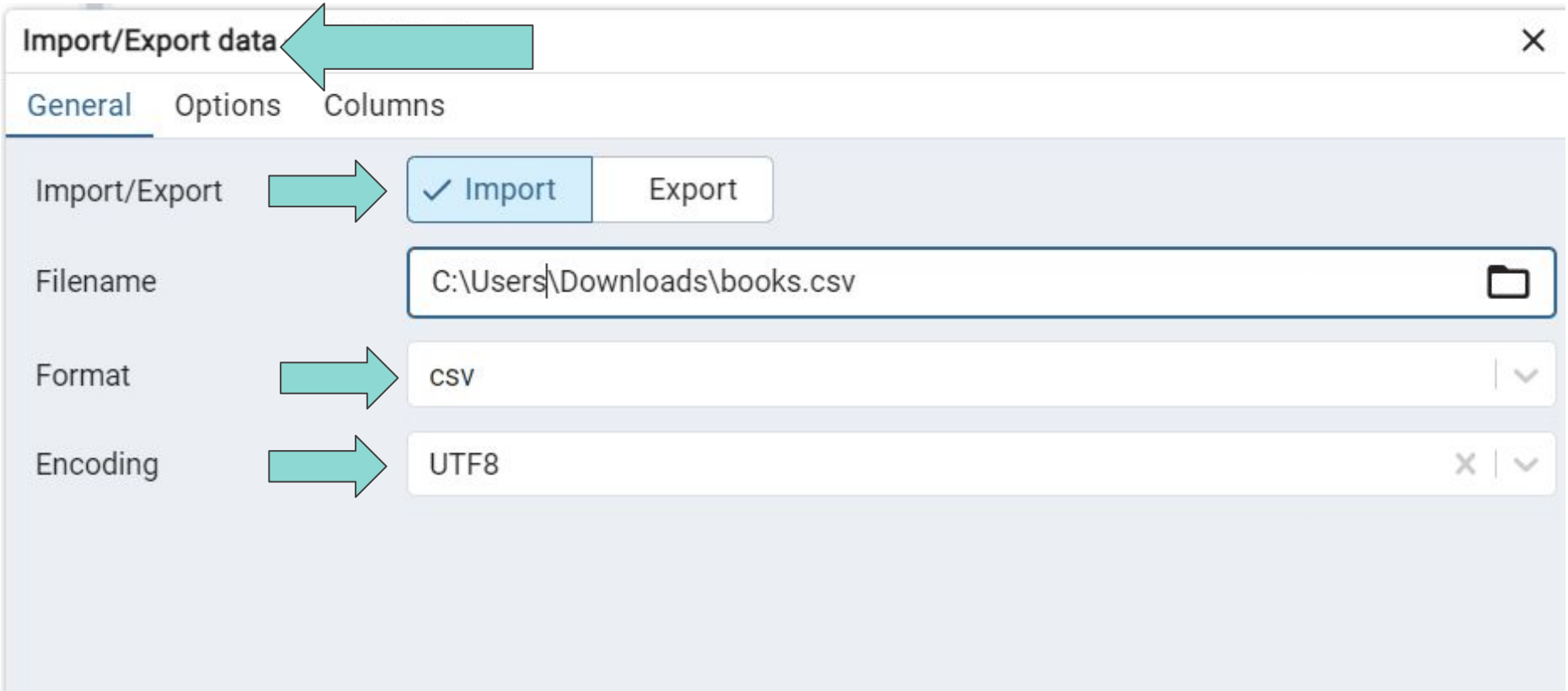
    first_name VARCHAR(50),
    last_name VARCHAR(50),
    experience VARCHAR(50),
    confidence INTEGER,
    study_field VARCHAR(100),
    job_title VARCHAR(100),
    strength CHAR(50)
);
```

Import- with UI

SELECT * FROM sql_students => empty table

Go to schema=> sql_students table=> right click=> import/export data=>import=>

Make sure the file you're importing is in *.csv* format.


Import/Export data
General
Options
Columns
Import/Export
Import
Export
Filename
C:\Users\Downloads\books.csv
Format
csv
Encoding
UTF8


Import/Export data - table 'airbnb'
General
Options
Columns
OID
Header
Delimiter
,
Comma
Specifies the character that separates columns within each row (line) of the file. The default is a tab character in text format, a comma in CSV format. This must be a single one-byte character. This option is not allowed when using binary format.
Quote
"
Specifies the quoting character to be used when a data value is quoted. The default is double-quote. This must be a single one-byte character. This option is allowed only when using CSV format.
Escape
'
Specifies the character that should appear before a data character that matches the QUOTE value. The default is the same as the QUOTE value (so that the quoting character is doubled if it appears in the data). This must be a single one-byte character. This option is allowed only when using CSV format.
NULL Strings
Specifies the string that represents a null value. The default is \N (backslash-N)
Close
Reset
OK

Import- with query

COPY student_info(first_name, last_name, experience, confidence, study_field, job_title, strength)

FROM 'C:\address\dataset-sql-students.csv'
DELIMITER ',' CSV HEADER;

Pay attention to giving “read permission” to the file!

Post-Import Validation:

- Run SELECT * FROM table_name;
- Use SELECT COUNT (*) FROM table_name; to verify row numbers
- Spot-check data quality (no misaligned columns)

Checklist:

- All rows were imported?
- Data types correctly recognized?
- Are there NULLs where there shouldn’t be?

Common Import Issues

❌ Data type mismatch
Fix: Edit column types or clean CSV data

❌ Header misalignment
Fix: Open CSV and rename headers to match table

❌ Encoding problems (é, ç, etc.)
Fix: Ensure CSVs are in UTF-8

❌ Empty columns or rows
Fix: Remove them before importing

CTRL + Space key:

Activate Autocomplete.

Importing 10 Datasets

- CREATE DATABASE datasets;

1. Iris Flower Dataset- Classic flower classification

The Iris dataset contains 150 flower samples from three species: Setosa, Versicolor, and Virginica. Each sample includes 4 features: petal length, petal width, sepal length, and sepal width (in cm).

Use Case:
Commonly used for practicing classification and data analysis techniques in machine learning and SQL.

Attributes:

- Sepal Length, Sepal Width, Petal Length, Petal Width, Species

2. Titanic dataset- passenger info+survival status

Passenger data from the Titanic’s tragic 1912 sinking. Out of 2,224 people on board, 1,502 lost their lives.

Use Case: Used to explore survival patterns based on age, gender, class, and more.

Attributes:
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

3. Country names and ISO codes

A complete list of countries with their ISO codes, useful for linking data to countries in visualizations or analysis.

Use Case:
Helps tools like Plotly or Power BI map data correctly using standard codes.

Attributes:

- Alpha-2 code (e.g. FR)
- Alpha-3 code (e.g. FRA)
- Numeric code
- ISO 3166-2 format

4. Netflix Movies and TV Shows

A list of Netflix movies and shows released between 2008–2021, covering various genres and countries.

Use Case:
Great for practicing SQL queries and data visualization with tools like PostgreSQL and Tableau.

Attributes:
Title, Type, Genre, Country, Release Year, Duration, etc.

ENUM (short for enumerated type) is a data type that lets you define a list of allowed values for a column.

If you want a column to only accept specific values like:

```
CREATE TYPE order_status AS ENUM ('Pending', 'Completed', 'Cancelled');
```

5. Books Dataset - Goodreads books info

A dataset of popular books collected via the Goodreads API, including ratings, reviews, authors, and publication info.

Use Case:
Ideal for book recommendations, ratings analysis, and exploring trends among readers.

Attributes:
Title, Author(s), Rating, Reviews, Publisher, Publication Date, etc.

6. NBA Player Stats

Detailed game-by-game stats for NBA players during the 2024/2025 season. Includes traditional and advanced metrics for in-depth analysis.

Use Case:
Ideal for performance tracking, predictive modeling, and player comparisons.

Key Columns:
Player, Team, Opponent, Date, Minutes, FG, 3P, FT, Rebounds, Assists, Steals, Blocks, Points, Game Score, etc.

7. Spotify Tracks Dataset

A dataset of tracks across 125 genres, including audio features like danceability, energy, tempo, and popularity.

Use Cases:
Music recommendation systems, Genre classification, Audio feature analysis

Key Columns:
Track ID, Artist, Track Name, Popularity, Danceability, Energy, Tempo, Valence, Genre, etc.

8. Airbnb NYC Listings

This dataset covers Airbnb listings, reviews, and availability in NYC, aimed at helping users practice data cleaning.

It includes added inconsistencies for hands-on experience with missing values, outliers, and formatting issues. Great for beginners learning data cleaning and analysis.

9. COVID-19

This dataset provides global COVID-19 data, including cases, deaths, recoveries, testing, and demographics.
Sourced from official health organizations, it's useful for tracking trends, research, and policy analysis.

10, 11. Google Play Apps info

This dataset, scraped from the Google Play Store, contains app details like category, rating, size, and more. It's valuable for analyzing app trends and guiding developers.

Primary key

- It gives each row a unique number, even if the other details are the same.

- Just like your credit card or library card number, it’s used to uniquely identify one thing in a table.

1. Decide which column(s) uniquely identifies each row (→ primary key).

2. Decide how two tables are related (→ foreign key).

Example

1. ALTER TABLE googleplay_apps ADD PRIMARY KEY (app);

1. ALTER TABLE googleplay_reviews ADD FOREIGN KEY (app) REFERENCES googleplay_apps(app);